6.2 Robotics

6.2.1 What is robotics?

The word **robot** comes from the Czech word ***robota*** (which means 'forced labour') and the term was first used in the 1920s play 'Rossum's Universal Robots'. The concept of the robot has fired the imagination of science fiction writers for countless years; indeed Isaac Asimov even composed his ***three laws of robotics***:

- a robot may not injure a human through action or inaction
- a robot must obey orders given by humans, unless it comes into conflict with law 1
- a robot must protect itself, unless this conflicts with law 1.

So what is a robot in the real world? **Robotics** is a branch of (computer) science that brings together the design, construction and operation of robots. Robots can be found in:



▲ **Figure 6.12** Robot welder

- factories
  - welding parts together
  - spray-painting panels on a car
  - fitting windscreens to cars
  - cutting out metal parts to a high precision
  - bottling and labelling plants
  - warehouses (automatic location of items)

- in the home
  - **autonomous** floor sweepers (see Figure 6.13)
  - autonomous lawn mower
  - ironing robots (for example, 'dressman')
  - pool cleaning
  - automatic window cleaners
  - entertainment ('friend' robots)



▲ **Figure 6.13** Robot carpet sweeper



▲ **Figure 6.14** Reconnaissance drone

- drones
  - unmanned aerial vehicles (UAVs) are drones that are either remotely controlled or totally autonomous using embedded systems
  - can be used in reconnaissance (for example, taking aerial photographs)
  - can be used to make parcel deliveries (for example, Amazon).

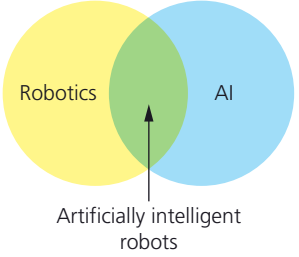
6.2.2 Characteristics of a robot

To be correctly called a robot, they need to have the following characteristics:

1 Ability to sense their surroundings:
   - this is done via sensors (such as light, pressure, temperature, acoustic, and so on)
   - sensors allow a robot to recognise its immediate environment and gives it the ability to determine things like size, shape or weight of an object, detect if something is hot or cold, and so on; all sensor data is sent to a microprocessor or computer.

2 Have a degree of movement:
   - they can make use of wheels, cogs, pistons, gears (etc.) to carry out functions such as turning, twisting, moving backwards/forwards, gripping or lifting
   - they are **mechanical structures** made up of many parts (for example, motors, hydraulic pipes, actuators and circuit boards)
   - they contain many **electrical components** to allow them to function
   - can make use of **end effectors** (different attachments to allow them to carry out specific tasks such as welding, spraying, cutting or lifting).

3 Programmable:
   - they have a 'brain' known as a **controller** that determines the action to be taken to perform a certain task (the controller relies on data sent from sensors or cameras, for example)
   - controllers are **programmable** to allow the robots to do certain tasks.



▲ **Figure 6.15** Robots and AI

It is important to realise that robotics and artificial intelligence (AI) are almost two entirely different fields:

TWO IMPORTANT NOTES:

1 Many robots don't possess artificial intelligence (AI) since they tend to do repetitive tasks rather than requiring adaptive human characteristics.

2 It is important not to confuse *physical robots* with *software robots* such as:
   - search engine ***bots*** or ***WebCrawlers*** (these 'robots' roam the internet scanning websites, categorising them for search purposes)
   - chat bots (these are programs that *pop up* on websites that seem to enter some form of conversation with the web user – see Section 6.3)

According to our definition above, software robots are not true robots.



▲ **Figure 6.16** Independent robot

Physical robots can be classified as **independent** or **dependent**:

» Independent robots:
   - have no direct human control (they are said to be **autonomous**, for example, an **autonomous** vehicle)
   - can replace the human activity totally (no human interaction is required for the robot to function fully).

» Dependent robots:
   - have a human who is interfacing directly with the robot (the human interface may be a computer or a control panel)
   - can supplement, rather than totally replace, the human activity (for example, in a car assembly plant where both humans and robots work together to produce a car).



▲ **Figure 6.17** Dependent robot

6.2.3 The role of robots and their advantages and disadvantages

We will now consider the use of robots in a number of areas, together with the advantages and disadvantages of using robots in each of these areas:

- industry
- transport
- agriculture
- medicine
- domestic (home) use
- entertainment.

Industry

Robots are used in many areas of industry, from heavy lifting work right through to delicate procedures. Examples include: paint spraying of car bodies, welding bodywork on cars, manufacturing of microchips, manufacturing electrical goods and automatic warehouses.

Control of robots is either through embedded (built-in) microprocessors or directly linked to a computer system. Programming of the robot to do a series of tasks is generally done in two ways:

1. The robot is programmed with a sequence of instructions which allow it to carry out the series of tasks (for example, spraying a car body with paint).
2. Alternatively, a human operator manually carries out the series of tasks; this can be done in two ways. In our example, we will assume an object is being painted using a robot arm.
   Figure 6.18 shows a robot arm equipped with a spray gun end-effector.
   Different **end-effectors** allow the robot arm to carry out many different tasks:
   - i the robot arm is guided manually by a worker when spraying the object; each movement of the arm is stored as an instruction on the computer; **or**
   - ii the worker straps sensors to his own arm and sprays the object; each movement is stored as a set of instructions on a computer; the sensors send back information such as position relative to the object, arm rotation and so on – this information forms part of the instructions stored on the computer.



▲ **Figure 6.18** Robot with spray gun end-effector

Whichever method is used, once the instructions have been saved, each series of tasks can then be carried out by a robot arm automatically. Each instruction will be carried out identically every time (for example, assembling parts in a television) giving a consistent product.

Robots are equipped with sensors so they can gather important information about their surroundings and also preventing them from doing 'stupid things'; for example, stopping a robot spraying a car if no car is present, or stop the spraying operation if the supply of paint has run out, and so on.

Robots are very good at repetitive tasks. However, if there are specialist tasks that require 'thinking' to cope with variable circumstances, (for example, making specialist glassware for some scientific work) then it is often better to still use human operators.

Table 6.1 shows the relevant advantages and disadvantages of using robots in industrial applications:

▼ **Table 6.1** Advantages and disadvantages of using robots

Advantages	Disadvantages
robots are capable of working in conditions that may be hazardous to humans	robots can find it difficult to do 'non-standard' tasks (for example, windscreen being fitted to a car is cracked)
robots work 24/7 without the need to stop	
robots are less expensive in the long run (since there will be fewer salaries to pay)	robots can lead to higher unemployment amongst manual labour tasks
robots are more productive than humans (higher productivity)	there is a risk of deskilling when robots take over certain tasks (for example, welding and paint spraying)
although not necessarily more accurate, robots are more consistent	
robots are better suited to boring, repetitive tasks than humans (therefore less likely to make mistakes)	factories can now be moved to anywhere in the world where operation costs are lower (leading again to unemployment in some countries)
there will be less cost in heating and lighting (robots don't need good light or warmth)	robots are expensive to buy and set up in the first place

Transport

Driverless vehicles are increasing in number every year. These are very complex robots, but the big problem is not really the technology (since problems will be ironed out through time), it is human perception. It will take a large leap of faith for humans to ride in a driverless car or an airplane with no pilot. We are already used to autonomous trains since these are used in many cities throughout the world. These systems have been generally accepted; but that is probably because trains don't overtake other trains and have a very specific track to follow (see notes later).

Autonomous cars and buses

In this section, we will consider autonomous cars as our example. Autonomous cars use sensors, cameras, actuators and microprocessors (together with very complex algorithms) to carry out their actions safely. Sensors (radar and ultrasonics) and cameras allow the control systems in cars to perform critical functions by sensing the dynamic conditions on a road. They act as the 'eyes' and 'ears' of the car.

Microprocessors process the data received from cameras and sensors and send signals to actuators to perform physical actions, such as:

- change gear
- apply the brakes
- turn the steering wheel.

Cameras catch visual data from the surroundings, while radar and ultrasonics allow the vehicle to build up a 3D image of its surroundings (very important when visibility is poor, such as heavy rain, fog or at night). Suppose an autonomous car is approaching a set of traffic lights that are showing red. The first thing is the control system in the car needs to recognise the road

sign and then check its database as to what action to take. Since the traffic light shows red, the microprocessor must send signals to actuators to apply brakes and put the gear into 'park'. Constant monitoring must take place until the light changes to green. When this happens, the microprocessor will again instruct actuators to put the car into first gear, release the brakes and operate the throttle (accelerator). This is a very complex set of operations since the microprocessor must constantly check all sensors and cameras to ensure moving off is safe (for example, has the car in front of it broken down or has a pedestrian started to cross the road, and so on). To go any further is outside the scope of this book.

Let us now consider some of the advantages and disadvantages specific to autonomous vehicles:

▼ **Table 6.2** Advantages and disadvantages of autonomous vehicles

Advantages of autonomous vehicles	Disadvantages of autonomous vehicles
safer since human error is removed leading to fewer accidents	very expensive system to set up in the first place (high technology requirements)
better for the environment since vehicles will operate more efficiently	the ever-present fear of hacking into the vehicle's control system
reduced traffic congestion (humans cause 'stop-and-go' traffic known as ***'the phantom traffic jam'***, autonomous vehicles will be better at smoothing out traffic flow reducing congestion in cities)	security and safety issues (software glitches could be catastrophic; software updates would need to be carefully controlled to avoid potential disasters)
increased lane capacity (research shows autonomous vehicles will increase lane capacity by 100% and increase average speeds by 20%, due to better braking and acceleration responses together with optimized distance between vehicles)	the need to make sure the system is well-maintained at all times; cameras need to be kept clean so that they don't give false results; sensors could fail to function in heavy snowfall or blizzard conditions (radar or ultrasonic signals could be deflected by heavy snow particles)
reduced travel times (for the reasons above) therefore less commuting time	driver and passenger reluctance to use the new technology
stress-free parking for motorists (the car will find car parking on its own and then self-park)	reduction in the need for taxis could lead to unemployment (imagine New York without its famous yellow cabs!)

Autonomous trains

As mentioned earlier, autonomous (driverless) trains have been around for a number of years in a number of large cities. As with other autonomous vehicles, driverless trains make considerable use of sensors, cameras, actuators and on-board computers/microprocessors. Autonomous trains make use of a system called **LiDaR** (Light Detection and Ranging); LiDaR uses lasers which build up a 3D image of the surroundings. Other sensors (such as proximity sensors on train doors) and cameras (including infrared cameras) are all used for various purposes to help control the train and maintain safety. The control system in the train also makes use of global positioning satellite (GPS) technology, which allows accurate changes in speed and direction to be calculated. Again, actuators pay a huge role here in controlling the train's speed, braking and the opening and closing of the train doors.



▲ **Figure 6.19** Autonomous train (London Transport)

Let us now consider some of the advantages and disadvantages specific to autonomous trains:

▼ **Table 6.3** Advantages and disadvantages of autonomous trains

Advantages of autonomous trains	Disadvantages of autonomous trains
this improves the punctuality of the trains	the ever-present fear of hacking into the vehicle's control system
reduced running costs (fewer staff are required)	system doesn't work well with very busy services (at the moment)
improves safety since human error is removed	high capital costs and operational costs initially (that is, buying the trains, expensive signalling and control equipment and the need to train staff)
minimises energy consumption since there is better control of speed and an optimum service requires less energy (trains stuck in stations still use energy)	ensuring passenger behaviour is acceptable particularly during busy times (for example, jamming doors open on trains, standing too near the edge of platforms and so on)
it is possible to increase the frequency of trains (automated systems allow for shorter times between trains)	passenger reluctance to use the new technology
it is easier to change train scheduling (for example, more trains during busier times)	no drivers mean there will be a need for CCTV to monitor railway stations

Autonomous (unpiloted) airplanes

Airplanes have used auto-pilots for many years to control flights. Human pilots only take over during take-off and landing. Autonomous (pilotless) airplanes would make even more extensive use of sensors, actuators and microprocessors to control all stages of the flight. Some of the main features of a control system on a pilotless airplane would include:

- sensors to detect turbulence to ensure smooth flights
- an increase in self-testing of all circuits and systems
- sensors that would automatically detect depressurisation in the cabin; thus allowing for quick stabilisation of the airplane
- use of GPS for navigation and speed calculations
- use of actuators to control, for example, throttle, flaps (on the wings) and the rudder.

Let us now consider some of the advantages and disadvantages specific to pilotless airplanes:

▼ **Table 6.4** Advantages and disadvantages of pilotless airplanes

Advantages of pilotless airplanes	Disadvantages of pilotless airplanes
improvement in passenger comfort (reasons given earlier)	security aspects if no pilots on-board (for example, handling terrorist attacks)
reduced running costs (fewer staff are required)	emergency situations during the flight may be difficult to deal with
improved safety (most crashes of airplanes have been attributed to pilot-induced errors)	hacking into the system (it might be possible to access flight controls)
improved aerodynamics at the front of the airplane since there would no longer be the need to include a cockpit for the pilots	passenger reluctance to use the new technology
	software glitches (recent software issues with modern airplanes have highlighted that software glitches can have devastating results)

Agriculture

With the world's population predicted to reach nine billion by the year 2050, more efficient agriculture via increased use of robotics is inevitable. Robots could replace slow, repetitive and dull tasks allowing farmers to concentrate on improving production yields. We will consider the following five areas where robotics could play a big role:

- harvesting/picking of vegetables and fruit
- weed control
- phenotyping (plant growth and health)
- seed-planting and fertiliser distribution
- autonomous labour-saving devices.

Harvesting and picking

- robots have been designed to do this labour-intensive work; they are more accurate (only pick ripe fruit, for example) and much faster at harvesting
- for the reasons above, this leads to higher yields and reduces waste (for example, **vegebot** (Cambridge University) uses cameras to scan, for example, a lettuce and decide whether or not it is ready to be harvested
- a second camera in **vegebot** (near the cutting blades) guides an arm to remove the lettuce from its stalk with no damage.

Weed control

- weed management robots can distinguish between a weed and crop using AI (see Section 6.3)
- examples of weed control robots are being used in France (by Mouton-Rothschild) to remove weeds between grape vines in their vineyards; this saves considerably on labour costs and improves vine growth
- weed control robots use GPS tracking to stay on course to move along the rows of vines and remove the weeds; a weed removal blade is operated by an actuator under the control of the controller (microprocessor) in the robot

- very often a **drone** (flying robot) is used first to do an aerial view of the vineyard, so that a programmed course of action can be produced, which is then sent to the weed control robot's memory.

Phenotyping

- **phenotyping** is the process of observing physical characteristics of a plant in order to assess its health and growth
- robots designed to do phenotyping are equipped with sensors (including spectral sensors and thermal cameras) that can create a 3D image/model of the plant, thus allowing it to be monitored for health and growth
- machine learning (see Section 6.3) is used to recognise any issues with leaves (for example, if they have a blight or have the wrong colour) so that the robot can convey this back to the farmer
- these robots are much more accurate and faster at predicting problems than when done manually.

Seed-planting drones and fertiliser distribution

- drones (flying robots) can produce an aerial image of a farm sending back a 'bird's eye view' of the crops and land
- they allow seed-planting to be done far more accurately
- they also allow for more efficient fertiliser-spreading to reduce waste and improve coverage (this is much more efficient than conventional crop spraying)
- drones can also be used in cloud seeding where the drone can add silver iodide crystals to a cloud forcing it to give up its rainwater
- the drones use a very complex camera system to target seeding and allow fertiliser spraying.

Autonomous agriculture devices

Several of the devices described above could be referred to as autonomous. The following list summarises some of the devices that can work independently of humans:

- grass mowers/cutters
- weeding, pruning and harvesting robots
- seeding robots
- fertiliser spraying
- all of these devices use sensors and cameras to go around obstacles, or they can even be programmed to 'go to sleep' if the weather turns bad.

Activity 6.8

Look through all the notes on use of robots in agriculture, and make a table showing all the advantages and disadvantages of using robots.

Activity 6.9

1 Describe three areas where robots can be used in agriculture to increase efficiency and reduce labour requirements.

For each example, write down the advantages and disadvantages of using robots.

2 Five statements are shown on the left and five computer terms are shown on the right.

By drawing lines, connect each description to its correct computer term.

no direct human control and has the ability to carry out human activity totally on its own	dependent robot
machine that requires human interface to operate and allow it to supplement human activities	actuator
branch of computer science that brings together the design, construction and operation of 'intelligent' mechanical machines	end-effector
devices attached to a robotic arm which allows it to carry out various tasks	autonomous
device, under the control of a microprocessor, which allows it to operate pumps, wheels, motors and pistons, for example	robotics

Medicine

- robots are used in surgical procedures, which makes the operation safer and also makes the procedures quicker and less costly
- robots can be used from monitoring patients to doing actual minor surgery
- the disinfecting of rooms and operating theatres can all be done by autonomous robots (similar to the types described in agriculture)
- robots can take blood samples from patients:
  - less painful to patients since the robot is better at determining a 'good vein'
  - safer to doctors and nurses if the patient has an infectious disease
  - doctors and nurses can be freed up to do more skilled work
- microbots can be used in target therapy:
  - these use microscopic mechanical components (including microprocessor) to localise a drug or other therapy to target a specific site causing less damage to surrounding tissue
- prosthetic limbs are now mini robots in their own right (since they meet the three characterisations of what defines a robot)
  - bionic skins and neural implants that interface with the human nervous system (of the damaged limb) giving feedback to allow for better control of the prosthetic limb (again sensors and actuators are used to give human-like responses, such as grip).

Domestic robots

Robots used around the house vary from devices to carry out household chores through to devices used to entertain people. For example:

- autonomous vacuum cleaners:
  - these use proximity sensors and cameras to avoid bumping into obstacles and allows them to cover a whole room automatically



▲ **Figure 6.20** Vector robot (personal assistant)

  - these robots have a microprocessor to control the overall operation of the device; this also allows the user to program the device
  - actuators are used to control motors which allow movement forward/ backward and from side to side
- autonomous grass cutters (mowers):
  - these use the same type of sensor, camera, microprocessor and actuator set up as vacuum cleaners
- personal assistants (such as 'Vector')
  - this is a robot controlled by a micro-processor that also uses cloud connectivity to connect to the internet
  - it understands voice commands (using a microphone) and will answer any questions it is asked
  - it also makes use of an HD camera, utilising computer vision, allowing it to recognise somebody's face as well as navigate a room (using proximity sensors and actuators) to steer around objects in its way.

Robots used in entertainment

The use of robots in the entertainment industry is increasing. They are now found in areas such as:

- entertainment parks and arenas/venues
- the film and TV industry.

The following examples indicate where robots are being used in the world of entertainment. The reader is advised to research the ever-increasing number of examples.

- theme parks are now using autonomous robots to entertain visitors to the park; these robots (often dressed as cartoon characters) can interact with visitors to allow them to engage safely with the theme park attractions and make the whole experience 'more realistic'
- music festivals are much more immersive for the audience; robotic methods are used to control lighting (including laser displays), visual effects and animation (e.g. superimposing an actor's image onto a robotic caricature and synchronising mouth movements); the visual performances can be fully synchronised with the music
- use of robots to control cameras; for example, keeping them steady and auto-focusing when moving around a scene; the movie *Gravity* used many robots to operate cameras, props and the actors (for example, to give an actor the appearance of moving around in the vacuum of space uncontrollably, robot arms were used to simulate human behaviour and produce life-like moving images)
- humanoid robots (either remote-controlled or pre-programmed) can perform 'stunt' action in movies/ television by performing tasks impossible for a human to do; they use CGI (computer-generated imagery) and image capture techniques to generate special effects
- robots are capable of producing special effects with a precision, speed and coordination which is beyond human capabilities; actions and special effects can be synchronised to within a millisecond and produce fully coordinated/ synchronised sound effects (e.g. movement of the mouth to match the sounds produced in a realistic manner).

Activity 6.10

Research five robots used as entertainment.

a Give the names of these five robots.

b Write down the advantages and disadvantages of the five robots you chose.

Activity 6.11

1 Write down **four** advantages and **four** disadvantages of using robots in the manufacturing industry.

2 To be referred to as a robot, it needs to be demonstrated that it has the following three characteristics:
- ability to sense its surroundings
- ability to move in some way
- a perceived intelligence.

For **each** of these **three** characteristics, describe **two** features you could use to demonstrate that a device could be called a robot.

3 Choose suitable words/phrases from the following list to correctly complete the paragraph that follows:

Word list:		
	activators	lasers
	actuators	light detection and ranging
	autonomous	primary vision
	cameras	radar and ultrasonics
	lane assist	sensors

Driverless cars are described as .....; these vehicles use ....., ..., microprocessors, software and ..... to allow them to function. A 3D image of the surroundings is produced by using ..... . Parking and nose to tail driving is achieved by using ..... in the bumpers.

4 Choose suitable words/phrases from the following list to correctly complete the paragraph that follows:

Word list:			
	actuators	environment	programs
	adaptive	intelligence	repetitive
	controller	microprocessor	sensors
	end-effectors	physical	system

Robots can collect data from their surroundings by using ..... . The data is then sent to a ..... to allow the robot to build up an image of its ..... Robots can perform various tasks by using different ..... . The 'brain' of the robot is often called a ..... which contains ..... to allow it carry out various tasks automatically. Many robots are not (artificially) intelligent, since they only do ..... tasks rather than requiring ..... human characteristics.

5 Autonomous robots are used in space exploration and in undersea exploration. These robots have to either work in the near vacuum of space or the very high-water pressures under the oceans. They need to be equipped with many sensors and cameras to carry out their remote tasks.

a Undersea robots are being used to investigate shipwrecks. Describe how the sensors and cameras could be used to photograph the shipwrecks. Also describe the role of the microprocessor and actuators in taking photographs and any samples needed from the shipwreck for further investigation.

b A space exploration robot has been sent on a mission to Mars. The robot needs to move around the surface of the planet safely

taking photographs and also taking soil/rock samples for later analysis.

i Describe how sensors, actuators and a microprocessor can be used to take samples from the planet's surface.

ii Describe three uses of the cameras on this autonomous robot.

c Describe the advantages and disadvantages of using autonomous robots in both undersea and outer space exploration.

d Give two other examples of where autonomous robots could be used.

6.3 Artificial intelligence (AI)

6.3.1 Introduction

Artificial intelligence (AI) is a branch of computer science dealing with the simulation of intelligent human behaviour by a computer. This is often referred to as the **cognitive** functions of the human brain (that is, the mental process of acquiring knowledge and understanding through thought, experience and the five senses). All of these cognitive functions can be replicated in a machine, and they can be measured against human benchmarks such as reasoning, speech and sight.

6.3.2 Characteristics of AI

Essentially, AI is really just a collection of rules and data, and the ability to reason, learn and adapt to external stimuli. AI can be split into three categories:

- narrow AI – this occurs when a machine has superior performance to a human when doing one specific task
- general AI – this occurs when a machine is similar (not superior) in its performance to a human doing a specific task
- strong AI – this occurs when a machine has superior performance to a human in many tasks.

Reasoning is the ability to draw reasoned conclusions based on given data/situations. Deductive reasoning is where a number of correct facts are built up to form a set of rules which can then be applied to other problems (for example, if AI is used to produce the perfect cup of tea based on a number of facts, the machine will learn from the experience and apply its new rules in the making of a cup of coffee, hot chocolate, and so on – modifying its methodology where necessary). By carrying out a sequence of steps, the AI machine can learn, and next time it will know how to do the task more effectively and even apply it to a novel/new situation. Thus the AI system is capable of learning and adapting to its surroundings. AI can very quickly discern patterns (which in some cases, humans cannot) and then make predictions by adapting to the new data. How all this is done is beyond the scope of this book (interested readers can find out more about this topic by consulting *Cambridge International AS and A Level Computer Science,* ISBN: 9781510457584).

Examples of AI include:

- news generation based on live news feeds
- smart home devices (such as Amazon Alexa, Google Now, Apple Siri and Microsoft Cortana):
  - the AI device interacts with a human by recognising verbal commands
  - it learns from its environment and the data it receives